Quantum Mechanics II — Exercise Sheet 7

Unitary evolution and moving packets

Variant: Questions only

Exercise 1 (A group with an unbounded generator). On $\ell^2(\mathbb{N})$ let $(U_t c)_n = e^{-\mathrm{i}tn^2} c_n$. (i) Show that this is a strongly continuous unitary group. For which c does $(U_t c - c)/t$ converge in ℓ^2 as $t \to 0$? (ii) Find the generator in the convention $U_t = e^{-\mathrm{i}tH/\hbar}$.

Exercise 2 (Locating the narrowest instant). A normalized Schwartz packet has initial variances $\Delta Q^2 = 9s^2$ and $\Delta P^2 = \hbar^2/s^2$, and symmetrized covariance $C_{QP} = -2\hbar$. Here $s > 0$ and $C_{QP} = \mathrm{Re}\langle (Q - \langle Q\rangle)\psi, (P - \langle P\rangle)\psi\rangle$. For free evolution find the time of least position variance and that variance. Check the covariance uncertainty inequality for the stated data.

Exercise 3 (A packet under a constant force). Let $F \in \mathbb{R}$ and $f \in \mathcal{S}(\mathbb{R})$ be normalized. Let χ_t be its free evolution. (i) Verify by differentiation that

$$\psi_t(x) = e^{\mathrm{i}(Ftx - F^2t^3/(6m))/\hbar} \chi_t\left(x - \frac{Ft^2}{2m}\right)$$

solves the Schrödinger equation with potential $V(x) = -Fx$ and initial vector f. (ii) Find its position and momentum means and its position variance in terms of the free packet. No assertion about a general self-adjoint domain for this Hamiltonian is required.

Exercise 4 (Time reversal of a free packet). For a normalized $f \in \mathcal{S}(\mathbb{R})$ let $Kf = \overline{f}$. (i) Show that $KU_t f = U_{-t} K f$ for free evolution. Relate the means of Q and P in Kf to those in f. (ii) Explain how conjugating a packet at its narrowest instant changes its later spreading.